3AL Tips

If you have any ideas for tips that could be added, please let me know ☺

Answering questions

You'll see a lot of questions of starting with the following phrases:

- "**True or false**". Your answer should be either "True" or "False" (not both, please). No explanation is necessary unless specifically asked for.
- "**Prove or disprove**". Either provide a proof or a specific counterexample that disproves the statement. It needs to be clear why your counterexample disproves the statement.
- "**Prove that for all X the property Y holds**". Proof by contradiction is usually a good idea. Assume that there is some object C that does not satisfy the property Y and try to get a contradiction from there.

Sets & maps

- To show that A is a **subset** of B (i.e. $A \subseteq B$), take an arbitrary element of a and show that it belongs to B (i.e. satisfies whatever necessary conditions needed to be an element of B).
- To show A **equals** B (i.e. $A = B$), you need to show $A \subseteq B$ and $B \subseteq A$.
- To show that $g : Y \to X$ is the **inverse** of $f : X \to Y$, show that $f(g(y)) = y$ and $g(f(x)) = x$ for all $x \in X$ and $y \in Y$.

Homomorphisms & isomorphisms

- To show that a map $\varphi : G \to G'$ is a **homomorphism**, show that it preserves operations.
- To show that a map $\varphi : G \to G'$ is an **epimorphism**, show that it preserves operations and is surjective.

- To show that a map $\varphi : G \to G'$ is a **monomorphism**, show that it preserves operations and is injective.

- To show that a map $\varphi : G \to G'$ is an **isomorphism**, show that it preserves operations, is surjective and injective. OR show that φ preserves operations *and* it has an inverse $\varphi^{-1} : G' \to G$ that also preserves operations.

- If G is a **group of order** 4 and you want to prove it is isomorphic to $\mathbb{Z}_4$ or K_4, determine whether G is cyclic. If it is cyclic, then it is isomorphic to $\mathbb{Z}_4$, otherwise it is isomorphic to K_4.

- If you want to show $G/H \cong G'$, use the First Isomorphism Theorem and define an *epi*morphism $\varphi : G \to G'$ where $\ker \varphi = H$.

- If you want to show G/H is isomorphic to a *subgroup* of G then define a homomorphism $\varphi : G \to G'$. Then the First Isomorphism Theorem says that $G/H \cong \varphi(G)$ (which we know is a subgroup of G').